

JEE MAIN REVISION MASTER: POLYMERS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 79 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Polymers - Core Application	79 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Polymers - Core Application (Very High Yield — 79 PYQs)

Core Principle: Key problem pattern in Polymers.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Match List I with List II : List I - (Monomer) List II - (Polymer) (A) Tetrafluoroethene (I) Orlon (B) Acrylonitrile (II) Natural rubber (C)..." [Ans: A]
- [2023 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Ethene at 333 to 343K and 6 - 7 atm pressure in the presence of $AlEt_3$ and $TiCl_4$..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Polymer used in orlon is :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Match List I with List II : List I - (Monomer) List II - (Polymer) (A) Tetrafluoroethene (I) Orlon (B) Acrylonitrile (II) Natural rubber (C) Caprolactam (III) Teflon (D) Isoprene (IV) Nylon-6 Choose the correct answer from the options given below :

- (A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) **[Correct]**
- (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- (D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Correct Answer: A

Solution Analysis: The correct match between List I (Monomer) and List II (Polymer) is as follows: (A) Tetrafluoroethene → (III) Teflon (B) Acrylonitrile → (I) Orlon (C) Caprolactam → (IV) Nylon-6 (D) Isoprene → (II) Natural rubber So, the correct answer is: A-III, B-I, C-IV, D-II